How Generous You Already Are Predicts Whether Fairness Gives or Takes

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***Abstract*—A group-fairness constraint requires two groups' selection rates to be close, but it does not say how the gap should close, and the certifying metric is content with either answer: it reads 0.018 where a demographic-parity constraint destroyed a fifth of the favourable decisions and 0.010 where it created more than it destroyed. A team cannot tell from its own dashboard which it has just done. We show the direction is set by the *population* rather than by which in-processing method is chosen, and is predictable before deployment from the rate at which the current model says yes, read against that population's own crossover — the rate a team already has, the crossover it must measure. Across 179 disjoint population samples spanning six decision domains, eight data sources and five countries, the constraint withdraws below a population's crossover and extends above it: on UCI Adult five in-processing methods each shrink the pool by 7.9–22.1%, while on mortgage lending the identical constraint grows it. Committed before ten never-measured populations, the rule scored 9 of 10 against a best constant of 6.**

The claim is narrow and sealed tests made it so. Direction is predictable within a population; magnitude is not, and below one percentage point the direction call is a coin flip. The practical output is an audit that locates a team's own crossover and *refuses* when it cannot: on fifty populations drawn at random, frame and seed fixed in advance, 78% admit a directional rule and 22% do not. That is the monotonicity gate alone; run end to end with every gate applied, the audit returns a direction on 29 of 52 stored sweeps — a little under six times in ten — and hands back the rest as unanswerable rather than scoring them. Two scope limits belong here: the relationship disappears under post-processing, and in five of the eight sources the favourable decisions are predicted labels rather than allocations actually made. Of twelve sealed or registered direction tests, one passed; we report the failures because each removed a stronger claim than the one that survives.

***Index Terms*—algorithmic fairness, levelling down, demographic parity, in-processing, selection rate, disparate impact**

I. Introduction

That a fairness constraint is silent about *how* the gap closes is well established. Mittelstadt *et al.* [11] made levelling down the centre of a well-known critique; Maheshwari *et al.* [13] observe that it "often goes unnoticed in the overall performance of the model"; Krco *et al.* [12] built an auditing framework precisely because an aggregate score reports neither how many people were affected nor in which direction; and Hu and Chen [6] show, on the dataset this paper opens with, that a parity-proxy constraint can leave both groups with fewer favourable classifications at once. We take all of that as given.

What the literature leaves open is the question a deploying team actually faces: **when** does each direction happen? The question is not academic, because the certifying number is not merely silent about the answer — it is equally content with either. On UCI Adult a demographic-parity constraint destroyed roughly a fifth of the favourable decisions, and the parity difference it was certified against read **0.018**. On HMDA mortgage data the identical constraint created more favourable decisions than it destroyed, and the parity difference read **0.010**. Both pass. A team looking only at its fairness dashboard cannot tell which of the two it has just done.

The claim, in one sentence. Whether a parity constraint hands out more favourable decisions or takes them away is set by the *population* being decided about rather than by which in-processing method is chosen — and a team can find out which, before deploying, by locating its own crossover and reading its current selection rate against it.

Two words in that sentence are doing work and we would rather name them than have them found. *In-processing*: under post-processing the relationship is absent, so the direction is not a population property in general, only within the family of methods studied here. *Locating*: the selection rate is a number every deploying organisation already has, and the crossover is not — it costs a sweep. The only version of this claim testable from a dashboard alone is a transported prior, whose record is 9 of 10 and then 5 of 10, and which we demote accordingly.

The audit that follows from it has *three* outcomes rather than two — the constraint will give, it will take, or the population admits no answer and the audit says so (Algorithm 1). The third is not a caveat attached to the method; it is a third of what the method does.

Four conditions bound the sentence above. Each is measured rather than assumed, and each was bought with a test that failed:

- **Direction only, and only above a floor.** A sealed model of magnitude lost to predicting zero (MAE 4.50 against 0.77). Effects below one percentage point are called correctly 61% of the time — a coin flip — against 95% above it.
- **About four populations in five.** On fifty populations drawn at random, frame and seed fixed before any arm ran, 78% admit a directional rule and 22% do not. The procedure *refuses* the 22% rather than guessing at them.

- **In-processing, not post-processing.** Under group-threshold post-processing [5] the relationship is absent ($r = -0.024$), and a sealed deconfounding test locates that boundary at the optimizer family rather than at access to the protected attribute (Section VII).
- **Mostly predicted labels, not allocations.** In five of the eight data sources the favourable decisions are model predictions rather than decisions actually made. HMDA, the one source recording real approvals, supports the direction claim on its natural arms but is also where our sweep protocol failed its held-out test.

The rule underneath is not surprising once stated: say yes often and the constraint tends to give, say yes rarely and it tends to take. What is *not* available without measuring is where the boundary falls for a particular population. Located crossovers span 0.22 to 0.81. In the one setting where we drew populations at random rather than choosing them — US state-level income prediction, fifty draws — no population's operating rate reached 0.55, and 25 of the 47 with a computable rate sat inside the 0.28–0.65 interval where most of them fall, which is precisely where an unaided reading of "often" against "rarely" has nothing to say. A further fifth of them follow no directional rule at all; there the intuition still returns a confident answer, and the audit declines to.

A. Scope, stated first

This is a paper about **attribute-blind in-processing** constraints, meaning the regime in which the protected attribute is unavailable at decision time. This is the regime most deployed systems occupy: in the United States, per-group decision thresholds face direct disparate-treatment exposure (*Ricci v. DeStefano* [27]; in credit, ECOA and Regulation B prohibit attribute-based terms outright), and commentators read the doctrine the same way [19], [26]. Two honest qualifications. First, the barrier is jurisdiction-specific, and the two major jurisdictions draw it in different places: in the US even *training-time* use of the attribute — which every in-processing method here involves — is legally unsettled, while in the EU the split runs the other way: *decision-time* use of special-category data faces GDPR Article 9 and direct-discrimination doctrine, but training-time use for bias detection and correction in high-risk systems is expressly permitted, under safeguards, by Article 10(5) of the AI Act — and for race and ethnicity the blind regime is often simply factual, because most member states collect no such attribute at all. Second, because of that split we motivate the regime as the one systems occupy, not as one any law cleanly compels. Under post-processing the effect we report disappears, and a sealed deconfounding test shows the disappearance belongs to the method, not to the attribute: the same in-processing optimizer, given the attribute, preserves the relationship (Section VII). The blind regime remains the paper's scope because it is where the record was built and where deployed systems sit — not because blindness is what the finding depends on.

B. Contributions

The phenomenon. An attribute-blind parity constraint withdraws favourable decisions on some populations and extends them on others, and the certifying metric reports the same kind of number in both cases: on UCI Adult five in-processing methods each shrink the pool by 7.9–22.1% at an exchange rate of 2.68 destroyed per created, while on HMDA mortgage lending the identical constraint grows it by 4.3% at 0.50. Measured over **179 disjoint population samples**, six domains, eight sources, five countries. Counted in people rather than rates the burden is more lopsided still, and that divergence is itself predictable ($r = +0.885$).

The audit, and its refusal. The same design lets a practitioner locate their own crossover from a deployed model, and return NO ANSWER when the population supports none. On fifty populations drawn at random — frame, seed and draw committed before any arm ran — 78% admit a directional rule and 22% do not, and the procedure hands back the latter rather than scoring them. Detecting a non-monotone landscape requires the sweep that transporting a prior exists to avoid, which is why the refusal cannot be replaced by a dashboard reading.

The predictor, identified by elimination rather than asserted. Holding the task fixed and moving only the decision rule separates the rate from task difficulty. A screen-gated Brazilian race cohort — ten arms from two person-samples, so the confound breaks *within* populations, though leaving either out drops the rule to 60% — beat a cutoff-only null 8–9 to 5; a sealed lending cohort beat a loan-purpose null 8 to 5 on real approvals; a five-domain comparison rules out the group gap. We say *predicts*, never *causes*: no pooled slope transfers.

A conditionality theory predicts, now measured. Contemporaneous theory [14] establishes distribution-dependence in the blind regime without experiments, its conditions being orderings on score regions rather than measurable quantities. A percentile-relaxed form of that ordering agrees with the measured direction on 24 of 26 arms and tracks our predictor at $r = +0.935$, holding across 150 arms at $r = +0.850$ (population-clustered 95% interval $+0.813$ to $+0.892$) and surviving a boosted-tree probe. The exact conditions were certified on none of the 26 by our plug-in estimator, whose ratio diverges in the $\hat{\nu}$ tail — an estimator pathology, and we tried no other.

Two predictive claims, and a prior demoted. A *sweep-conditional* claim — measure your own crossover, then read your rate against it — and a weaker *transported-prior* claim, the only one testable before any measurement. The rule itself is sealed: committed before ten never-measured populations, it scored 9 of 10 against a best constant of 6. The fixed *value* 0.54 fares worse and we no longer present it as a finding: located crossovers span 0.22 to 0.81, the value failed its own strict bar on the race cohort, and on a second post-hoc cohort it scored 5 of 10 — each miss agreeing with that population's own sweep.

Boundaries stated rather than left for review to find, including a computable test for when the procedure cannot be used at all. Of twelve sealed or registered direction tests, one passed; each failure removed a stronger claim than the one that survives, and one cohort built to strengthen the paired statistic showed the attempt self-defeating by construction. The relationship survives changes of route, tolerance, base learner and protected attribute, and disappears under post-processing.

II. Related Work

What we take as given. Levelling down as a phenomenon and minimum-rate constraints as a remedy are Mittelstadt *et al.*'s [11]; the impact decomposition we use throughout is Krco *et al.*'s [12]; fairness gerrymandering is Kearns *et al.*'s [15], which we replicate with a boundary attached (Section VI); the income-cutoff knob is Ding *et al.*'s [16]; the reductions we constrain with are [1], and we benchmark against minimax group fairness [20] rather than only against the unconstrained reduction. Contemporaneous theory [14] establishes that in the blind regime the direction is distribution-dependent, so we do not claim the conditionality itself — what it contains is no experiments and conditions stated as orderings on score regions rather than measurable quantities.

Four adjacent results, and what each leaves open. Hu and Chen [6] come closest to the phenomenon: an in-processing parity-proxy constraint on *this paper's own dataset* leaving both groups with fewer favourable classifications, read as a Pareto violation. Their existence claim is prior to ours and we do not repeat it as new; what they do not do is parameterise the effect, and their conclusion — that welfare moves non-monotonically in the fairness tolerance within one population — concerns a different axis from ours, which compares unconstrained against constrained across populations. Liu *et al.* [7] is the closest structural analogue: a critical selection rate at which the sign of a parity constraint's effect flips, with the baseline's position relative to it doing the predicting. Three things differ and the resemblance is worth stating plainly rather than leaving for a reader to find — their flipping quantity is one group's expected score change over a time step against our one-shot headcount over both groups, their axis is the protected group's own rate against our model's overall rate, and their policies are group-specific where ours are blind. Menon and Williamson [8] give the parity-optimal blind classifier as a per-instance threshold correction, pushing some subjects out of the positive set and pulling others in without evaluating the net. The infra-marginal line [9], [19] supplies the vocabulary a mechanism would need — what happens at the margin, not away from it — and a direction flip of its own, though theirs is one group's share of a fixed budget rather than a pool that can change size.

On not explaining the mechanism. Our sealed derivation lost to a constant (Section X) and we claim no mechanism, but we should not overstate that into an absence of any account. For the attribute-*aware* Bayes-optimal case the parity-optimal thresholds are available in closed form [10] as opposite-signed shifts each inversely proportional to a group's size; to first order the group sizes cancel and the sign of the pool change is set by which group carries more score density at the decision margin — an infra-marginality statement, and a property of the population rather than of the method, which is what we measure. We could not extend it: our constraint is blind, our classifier is a reduction's randomized mixture rather than a threshold rule, and the effects we measure are far outside a first-order regime. The honest statement is that we failed to extend a known marginal argument to the blind randomized case, not that no account exists.

III. Setup and Method

Data. 179 independent populations across eight sources. Two are US surveys: UCI Adult (45,222 rows) and ACS Income [16] over 102 state-years and two protected attributes. One records real allocations: HMDA 2018 mortgage filings, 66 rate-bearing arms over 40 states. The rest are IPUMS International census extracts for Brazil and Mexico, COMPAS [18], LSAC bar passage, the Dutch national census of 2001, and Taiwanese credit-card default records. The last three were added to leave the sources the finding was formed in, and chosen for what they test: the Dutch census carries a between-group gap of 0.298, roughly twice Adult's, which makes it the hardest case for the standard "it is really the group gap" alternative; LSAC was chosen because bar passage is naturally generous (base rate 0.890) and should therefore be unsweepable.

What a favourable decision is, and mostly is not. This belongs here because it bounds everything after it. **In five of the eight sources no one receives anything**: the favourable decisions are a classifier's labels over people whose incomes, occupations or bar results already exist, so the pool being given and taken is a pool of predictions. HMDA is the one source recording real allocative decisions — and it is where the direction reverses, and where our sweep protocol failed its held-out test. Whether the construct transports from status-prediction benchmarks to allocation data is an open question, not an assumption we are entitled to. On HMDA the outcome is the *lender's* approval: action-taken codes 1 and 2 against 3, with withdrawn, incomplete, purchased and preapproval records excluded as not being a completed approve-or-deny, and features allow-listed to what exists when the application is filed.

Definitions. An *arm* is one population under one configuration — a decision threshold, a label cutoff, a tolerance, a learner. The *selection rate* r is the fraction of the test split the baseline marks favourable. The *pool change* Δpool is the percentage change, baseline to mitigated, in the number of favourable decisions; *extension* and *withdrawal* name its sign. A population's *crossover* is the rate at which that sign changes, reported as the bracket between the highest rate that still shrinks the pool and the lowest that grows it. The *viable band* is the span of rates over which the baseline stays above the trivial predictor. The *exchange rate* is favourable decisions destroyed per favourable decision created, an unweighted descriptive count.

Protocol and convention. An L2-regularised logistic regression on a 70/30 split, five seeds per arm, with every reported figure a seed mean; the constraint is Fairlearn's exponentiated-gradient reduction under demographic parity [1] at a tolerance of 0.01 unless stated. $y = 1$ *is always the favourable outcome.* That required inverting COMPAS's recorded target, which encodes recidivism, and declaring *Female* the advantaged group on its sex arm, since women in that cohort reoffend less. Both are enforced by automated checks, because either error produces every directional result sign-inverted without raising.

Exclusion rules, and where they bite. An arm is discarded if the baseline parity gap is below 0.05, or if the baseline accuracy falls below $\max(p, 1-p)$, the accuracy of always predicting the more common label. Both were developed *during* the exploratory phase in response to failures the ledger records, so exploratory correlations are conditioned on in-sample-tuned filters and Table VIII reports the dependence. What the sealed tests did with them varies, and stating it as uniform would flatter them: the re-seal applies *neither* and scores all ten populations raw; the third-direction and lending cohorts apply only the magnitude guard their own protocols fix; the screen-gated race cohort is the one carrying both, frozen before its extract existed. The sealed record is therefore less filtered than an unqualified claim of frozen guards would suggest, which is the direction that favours it.

Three survey conventions, because every crossover here depends on them. Inherited from the folktables benchmark [16]: ACS records are used unweighted, incomes are nominal without the inflation adjustment, and household clustering is not modelled. Every selection rate is therefore a property of the unweighted sample rather than a design-consistent state estimate, and the numeric locations — including the 0.54 prior this paper argues about — are properties of that convention. The weighted re-analysis has run: under person weights **every located crossover shifts down by 0.03–0.08**, without changing any population's ordering or any direction call. That shift is the same order as the 0.54-against-0.50 distinction of Section IX-C, which is a reason to read the *ordering* as the finding and the value as a convention-relative landmark. Seed-resampled intervals are anti-conservative for the same reason.

Accounting. A *population* is one unit — a state, cohort or census — and *independent* means disjoint person samples: no person appears in two, and arms within a population share people, so however many arms a population contributes it is counted once. Ratios here are over populations unless they say otherwise. Table III states which populations enter which analysis; the per-ratio denominators are computed from the stored results rather than maintained by hand, since four counts in this project's own history turned out to be arm counts masquerading as population counts.

TABLE I

WHICH POPULATIONS ENTER WHICH ANALYSIS, FOR THE ANALYSES THIS PAPER REPORTS. THE COMPLETE RECORD CARRIES FURTHER ROWS FOR RESULTS THAT ARE CITED HERE BUT PRESENTED THERE.

is one population under one configuration; populations are counted once however many arms they contribute.

Analysis	**Populations (arms)**
Ablation; who pays	Adult (5 methods $\times$ 5 seeds)
Rate–people divergence	10 pops. (19 arms, 2 attributes)
Label route (income cutoffs)	AL sex; AL, MS, OR, UT race (4 each)
Natural transition	HMDA MS+LA (5 loan purposes)
Threshold route, dense sweeps	Dutch, SC, COMPAS, AL, KY (12 pts.)
Tolerance, 25$\times$ range	AL, CT, KY, OR, SC
Boosted trees	the same five states
Post-processing regime	17 populations ($r = -0.024$)
Attribute-aware direction	27 of 27, of which 9 pre-registered
Relaxed ζ ordering	26 arms from 15 populations
Crossover intervals	4 non-lending + 3 lending estimates
Sealed test 1 (failed)	8 scored + Taiwan, all fresh
Re-seal (holds)	10 fresh ACS states
Sealed magnitude (failed)	NY + TX (16 arms)
Selection-rate floor	10 pops. (19 arms)
Third direction cohort (failed)	24 fresh state-years, 2014 + 2019
Sealed lending cohort (failed)	8 fresh markets (16 arms, 2 purposes each)
Lending coverage	40 race + 26 sex arms (66)
Landscape survey	50 pops. at random (660 arms)

TABLE II

FIVE IN-PROCESSING METHODS ON UCI ADULT, FIVE SEEDS: THE TWO REDUCTIONS AND GRIDSEARCH FROM [1], ADVERSARIAL DEBIASING AFTER [2], AND THE PREJUDICE REMOVER AFTER [3]; DI IS THE DISPARATE-IMPACT RATIO OF [4]. EVERY ONE REACHES PARITY, AND EVERY ONE SHRINKS THE POOL.

Method	**Acc.**	**DP**	**EO**	**DI**	**Δpool**
Unconstrained	0.847	0.186	0.095	0.300	—
ExpGrad (DP)	0.828	**0.018**	0.280	0.895	-20.5%
GridSearch (DP)	0.827	**0.015**	0.304	0.986	-22.1%
Adversarial deb.	0.826	0.020	0.250	0.889	-14.8%
Prejudice remover	0.837	0.065	0.188	0.686	-10.2%
ExpGrad (EO)	0.836	0.107	**0.032**	0.521	-7.9%

IV. PARITY IS REACHED BY WITHDRAWAL — IN SOME POPULATIONS

Table II gives the ablation. Every method reaches its target: demographic parity falls from 0.186 to as little as 0.015 for under two accuracy points, and this holds on a linear model and on an adversarial one alike. The mitigation itself works, which is why we treat it as background rather than as a contribution.

What changes alongside is the part the metric does not report: **every one of these methods shrinks the pool**, by 7.9% to 22.1%, and not one of them closes the gap primarily by extending decisions to the disadvantaged group. Eighteen of nineteen survey arms show the same pattern, which suggests this is a property of the populations rather than of any one method.

A. Rates understate what is happening to people

The decomposition of [12] splits the closed gap into the part paid by the advantaged group losing ground and the part

TABLE III
WHO PAYS, UCI ADULT. THE RATE-LEVEL VIEW AND THE PEOPLE-LEVEL VIEW OF THE SAME INTERVENTION DISAGREE SYSTEMATICALLY.

Method	Share (rates)	Share (people)	Exchange
ExpGrad (DP)	0.575	**0.742**	2.68
GridSearch (DP)	0.575	**0.743**	2.69
Adversarial deb.	0.508	**0.700**	1.96
Prejudice remover	0.498	**0.673**	2.04
ExpGrad (EO)	0.531	**0.660**	1.92

gained by the disadvantaged group. Measured in *rates*, that split is close to even — 0.50 to 0.58 of the closure comes from withdrawal (Table III).

Measured in *people* the split is not even. Between 0.66 and 0.74 of the individuals moved in the two directions that close the gap — advantaged subjects losing a favourable decision, plus disadvantaged subjects gaining one — are advantaged subjects losing one. The denominator is those two flows and not every changed decision: it excludes the smaller counter-flows in each group, and a reader who takes it for all churn will find it inconsistent with the exchange rate, which is counted over every flow. The reason the two views disagree is that equal-looking rate changes fall on groups of unequal size, so the rate-level view — which is what a fairness dashboard reports — understates the burden counted in individuals, and it does so in every method tested. The divergence between the two views is itself predictable from a cross-flow diagnostic, at $r = +0.885$ across populations.

The **exchange rate** shows the same thing without needing a share: ExpGrad under demographic parity destroys 2.68 favourable decisions for every one it creates.

The normative baseline, stated as the convention it is. Throughout, "withdrawal", "who pays" and the exchange rate are counterfactual-comparative quantities measured against the *unconstrained model's* allocation — a baseline with no privileged moral standing, since the constraint exists precisely because that allocation is suspect. On a claims-based view [28], withdrawing an approval that answered to no legitimate claim wrongs no one; a positive classification is also not automatically a benefit (extension in lending can be predatory inclusion). The exchange rate is therefore an unweighted descriptive count, not a welfare ledger: a reader with prioritarian weights on the disadvantaged group's gains may score a rate above 1.0 as net-good, and the group-conditional flows the tables report make such weighted readings computable. What the accounting establishes is narrower and metric-relative: the certifying number cannot tell these outcomes apart, whatever one's weights.

On HMDA mortgage data, however, the direction reverses. The identical constraint *grows* the pool by 4.3%, at an exchange rate of 0.50 — two favourable decisions created for every one destroyed. What makes this worth noticing is that the parity metric cannot see the difference: it reports 0.018 on the population that destroyed a fifth of its favourable decisions, and 0.010 on the one that created more than it destroyed.

V. WHAT PREDICTS THE DIRECTION

A. A single-factor design

ACS Income's label is "earns more than \$50,000", and that threshold is a benchmark convention rather than a fact about the world, which makes it a knob we can turn. Moving it on one fixed state varies the selection rate while the sampled people, instrument, feature set and group ratio stay fixed — though moving the label redefines the prediction problem itself, so this is one arm of the two-routes triangulation below, not an isolation of the rate on its own. The change in the pool rises with the baseline selection rate at $r = +0.801$ over four arms — too few points for formal inference, and reported as descriptive; a partial correlation on four points has one degree of freedom and is omitted as uninformative. The sign flip is the substantive observation: 22 decisions are destroyed per one created at a rate of 0.03, compared with 0.75 at 0.89.

B. The transition appears without being manufactured

The same transition also appears where nothing is manipulated at all. Pooling two states across five real loan products, home improvement at a selection rate of 0.555 levels *down* (-1.57%) while refinancing at 0.871 levels *up* ($+2.95\%$), with $r = +0.803$ and Spearman $\rho = +0.900$. This means a bank running one constraint across its loan book would take opportunities away in one product while handing them out in another, and its fairness report would show success in both.

C. It is the rate, not the difficulty of the task

The design above moves the *label*. "Earns over \$100,000" is not a rarer version of "earns over \$20,000"; it is a harder problem. That design therefore cannot separate two explanations which predict everything observed so far.

We separate them by holding the task completely fixed — same rows, features, groups and *fitted scores* — and moving only where the model draws its line. The relationship reproduces, and so does the crossover's location: on Oregon the two routes give 0.362–0.653 and 0.353–0.637 (Fig. 5). **Task difficulty is not doing the work.**

With twelve points chosen from each population's viable band, the relationship holds on the Dutch census (+0.946), South Carolina (+0.905) and COMPAS (+0.844). Alabama and Kentucky return *negative* values on eleven and ten arms; their viable bands top out at 0.566 and 0.561, at or below every crossover measured, so both sweeps lie almost entirely on one side of the transition. **We therefore claim the relationship across the crossover and withdraw any claim that it holds within the low-rate region alone.**

D. Five domains, and the group-gap alternative

Fig. 1 shows every arm, and Table IV the summary. The table's job is *breadth*, and its two halves carry different weight. The four- and five-arm rows are descriptive, exactly as Section V-C concedes for Alabama. The dense-sweep columns carry more, but not as much as a Pearson test would suggest: twelve thresholds applied to one fitted score vector over one sample are not twelve independent observations, so we quote

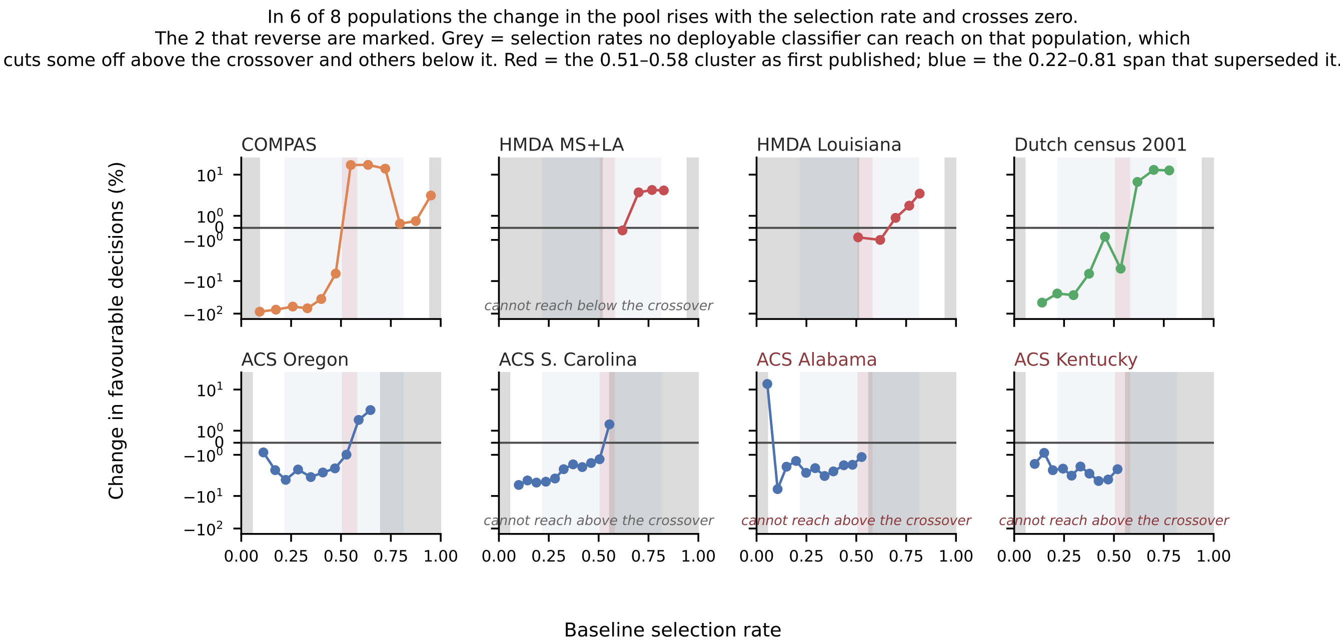

Fig. 1. **Within a population**, the change in the pool rises with the baseline selection rate and crosses zero. Eight densely swept populations, every retained arm; the two that reverse are marked, and both are sweeps whose viable band confines them below the crossover. Panels are deliberately *not* pooled — no pooled slope survives its pre-registered bar, so one scatter would assert a common curve the data reject. These eight are 5% of the record and are shown for shape, not for scale; Figures 2 to 3 carry the population-level evidence.

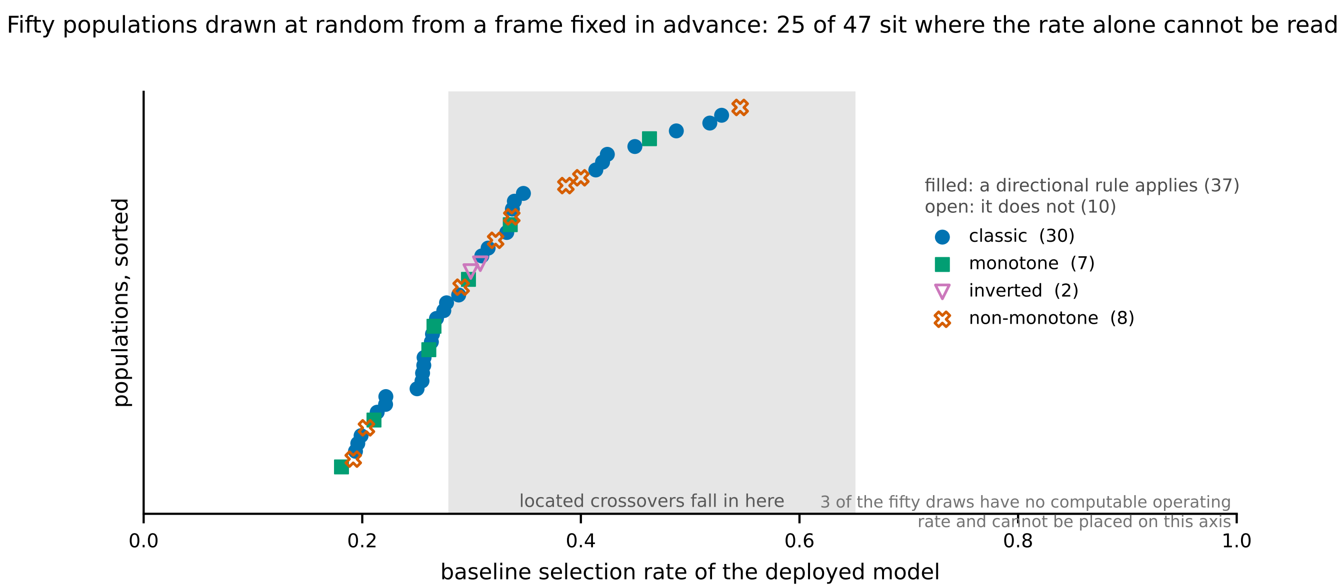

Fig. 2. **The scope of the claim, on a sample nobody chose.** All fifty populations of the landscape survey — frame, seed and draw committed to version control before any arm ran — placed at the baseline selection rate of their deployed model and coloured by the verdict the audit returns. 78% admit a directional rule and 22% do not, and the non-monotone populations are scattered through the range rather than confined to one end, which is why no rate threshold can screen for them. Note also what the sample does *not* contain: no population reaches a rate of 0.55, and 25 of the 47 with a computable rate sit inside the band where located crossovers fall, where reading "often" against "rarely" unaided has nothing to grip.

no p-value on them — such a test describes whether a sweep is monotone, not whether the rate predicts across populations. The pre-registered alternative — "this is a property of income prediction and American mortgage lending", requiring $|r| < 0.30$ — is refuted on both new instruments, and refuted on their *densely swept* arms rather than their sparse ones: COMPAS gives $+0.844$ and the Dutch census $+0.946$ over twelve points each. Earlier versions of this paper attached

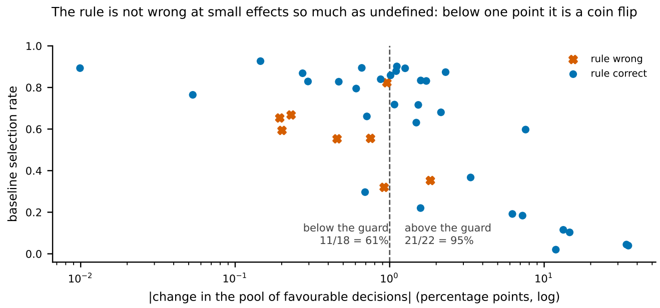

Fig. 3. **Where the rule is undefined rather than wrong.** All forty arms of the two sealed direction cohorts, by the size of the effect they were asked to call. Almost every miss sits left of the one-point guard: above it the rule is correct on 21 of 22, below it on 11 of 18, which is a coin flip. The misses are concentrated rather than scattered, and they are not concentrated at any particular selection rate — it is the magnitude that governs, not where the population sits. What this measures is the guard's *existence*; its value is a round number fixed in a protocol, and every floor from 0.25 to 5.00 separates.

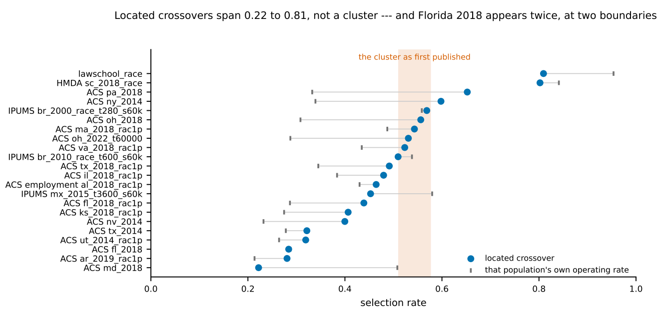

Fig. 4. **Every located crossover, against the cluster we first published.** Each population's measured crossover (dot) and its own deployed operating rate (tick). The shaded band is the four-population cluster of the first version of this work; the measured span is 0.22 to 0.81, so the cluster was an artefact of which populations had been swept. Florida 2018 appears twice — read by sex it crosses at 0.284, read by race at 0.439 — so a crossover belongs to a (population, attribute) pair and there is nothing here to look up instead of measuring.

TABLE IV
ONE POPULATION PER INSTRUMENT, ON WHATEVER ARMS THAT POPULATION NATURALLY SUPPLIES. CORRELATIONS ARE BETWEEN THE BASELINE SELECTION RATE AND THE CHANGE IN THE POOL; n IS THE NUMBER OF ARMS. NOT POOLED ACROSS INSTRUMENTS, FOR THE REASON FIG. 1 IS NOT POOLED.

			natural arms		**dense sweep**	
Instrument	**Domain**	**Population**	r	**n**	r	**n**
ACS Income	income	Alabama	$+0.801$	4	—	—
ACS Income	income	S. Carolina	—	—	$+0.905$	12
HMDA	mortgages	MS+LA pooled	$+0.858$	4	—	—
COMPAS	crim. justice	race arm	$+0.870$	5	$+0.844$	12
Dutch census	occupation	sex arm	$+0.915$	6	$+0.946$	12
LSAC	legal educ.	—	unsweepable	—	—	—

$p < 0.001$ to those coefficients; we withdraw it. Twelve thresholds on one fitted score vector over one sample are near-deterministic transformations of each other, so the Pearson null is the wrong reference and the effective sample size for a claim about *populations* is one. The population-level statistic this design supports is much weaker: across the six densely swept populations the within-population slope is positive in four and negative in two, a sign test at $p = 0.34$. We report it because it is the honest number, and note that the two negatives are the sweeps Section V-C withdraws for sitting entirely below their crossover — which is a stated exclusion, not a rescue, and either way the prospective evidence for this paper lives in the sealed forecasts and not in these coefficients. The four- and five-arm rows are reported for breadth and carry no inferential weight on their own.

The oldest competing account is that the *gap between groups* drives the direction. The Dutch census has a gap of 0.298, roughly twice Adult's, making it the population where that explanation should win. It gives $r = +0.915$ across all six arms, none excluded.

Because the parity-gap floor was tuned in-sample (Section III), the *natural-arm* column of Table IV was re-scored at floors 0.02/0.05/0.08/0.10: each of those correlations stays between $+0.80$ and $+0.92$ with no sign change at any floor (HMDA is floor-invariant outright; Alabama thins to two arms only at 0.10). **The dense-sweep column is a different matter, and an earlier version of this paper claimed otherwise.** Table VIII's 0.05 column *is* that dense-sweep column, and one of its three entries is strongly floor-dependent: South Carolina reads $+0.905$ at the committed floor and $+0.012$ at 0.02 and 0.01. So the floor is doing real work on a number this paper leans on, and the correct statement is the narrower one — the natural-arm evidence is floor-robust, the dense-sweep evidence is not uniformly so, and South Carolina's twelve-point correlation should be read as conditional on the exclusion rule rather than as independent of it.

TABLE V
WHAT EACH MANIPULATION PRESERVES

Manipulation	**Direction**	**Magnitude**
Route to the rate	unchanged	differs $3\times$
Tolerance, $25\times$ range	unchanged	differs $4\times$
Boosted trees (5/5 pops.)	unchanged	differs $3\times$
Attribute, sex $\to$ race	unchanged	—
Criterion $\to$ equalized odds	weaker, unevenly	differs $8\times$
Optimiser $\to$ post-proc.	**vanishes**	—

E. Robustness

Table V summarises. **Within a population, and in the attribute-blind regime, the selection rate predicts which way and orders how much. Neither transfers between populations**: the signed distance from a population's own crossover gives ρ of $+0.96$, $+0.95$ and $+0.78$ in three of four populations, but a pooled slope reaches only $r = +0.487$ against a pre-registered $+0.70$ bar.

VI. A CONSTRAINT ON ONE ATTRIBUTE LEAVES SUBGROUPS BEHIND

A constraint is imposed on the attribute someone chose to name. Table VI is the gerrymandering failure of Kearns *et al.* [15] measured on Adult.

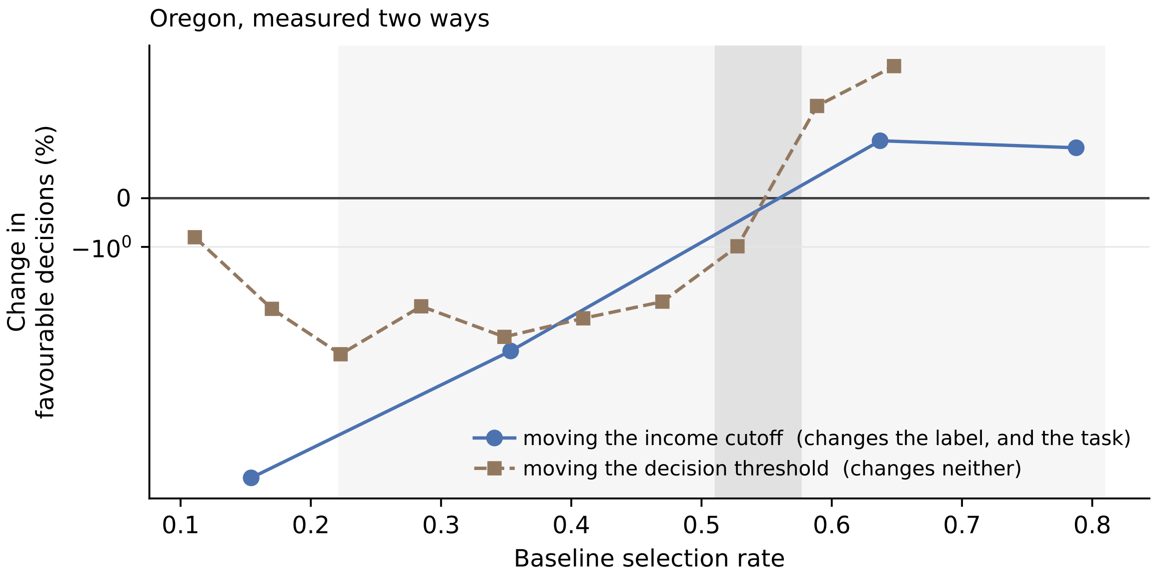

Fig. 5. The design that separates the selection rate from the difficulty of the task. Moving the income cutoff changes the label, and therefore how hard the problem is; moving the decision threshold changes neither, only where the fitted model draws its line. Both routes cross zero inside the same band. If difficulty were doing the work, the second curve would not reproduce the first.

TABLE VI
UCI ADULT. CONSTRAINING SEX TO 0.020 LEAVES THE WORST *Sex×Race* SUBGROUP GAP AT 0.178 — NINE TIMES LARGER — AND MOVES WHICH SUBGROUP IS WORST OFF.

Arm	Sex	Subgroup	Worst off
Unconstrained	0.190	0.315	F×Black
Constrained on sex	**0.020**	**0.178**	M×Black
Constrained on both	0.016	0.048	F×White

Constraining sex drives its gap from 0.190 to 0.020 — a success by the certifying metric. The worst *Sex×Race* subgroup gap falls only from 0.315 to 0.178, which is **nine times** the gap that was constrained. The subgroup carrying the residual also *changes*, from Female × Black to Male × Black, so an auditor tracking the originally worst-off cell would see improvement while a different cell absorbed the cost.

This replicates across populations and is more severe than Adult suggests: 13.2× in Mississippi. It is bounded, however, by how large the minority is — the effect scales with minority share at $r = -0.671$, and does not appear at all below a threshold. We present this as an empirical replication of [15] and [13] with a boundary attached, not as a new observation.

A measurement hazard worth naming. Roughly five subgroups per population are too small to support any estimate. A rate with an empty denominator must return undefined rather than zero, or an unmeasurable subgroup masquerades as a perfectly fair one — which is the failure this analysis is most likely to produce and hardest to notice.

VII. THE REGIME BOUNDARY

The relationship stops where the optimizer family changes, not where the protected attribute becomes readable, and a sealed test with both readings' bars committed in advance is what separates those. Under group-threshold post-processing [5], at each population's natural operating point, the rate carries no information about the pool's direction ($r = -0.024$, against $+0.585$ for the reduction on the same populations); our pre-registered prediction that the rule would carry over is a recorded failure. The missing cell decides between the two available explanations: given the attribute under the *same* in-processing optimizer, the relationship survives at $r = +0.672$ across thirteen natural arms — the method reading, not the blindness reading. Two cautions we would rather state than have computed for us: at thirteen arms the interval on that coefficient is wide enough to touch the competing reading's region, and its difference from $+0.585$ is not itself a difference. Neither coefficient carries much on its own; what the regime result rests on is the positive cell, and on the reduction retaining the relationship where the post-processor does not.

The theory's group-level claim is untouched. Where the attribute is readable the advantaged group loses and the disadvantaged gains, without exception across our attribute-aware arms. That is a consistency check against a published theorem rather than a forecast — the theorem made the prior close to one — and it separates cleanly from the pool question: the theorem fixes each *group's* direction wherever the attribute is readable, while the *pool's* direction, the sum it does not constrain, stays rate-predictable under in-processing in both regimes and unpredictable only under post-processing. A percentile-relaxed form of the theory's ordering agrees with the measured direction on 24 of 26 arms and tracks our predictor at $r = +0.935$, so the selection rate is a cheap proxy for a quantity the theory states expensively; the exact conditions were certified on none of the 26 by our plug-in estimator, whose ratio diverges in the $\hat{\nu}$ tail, and we tried no other.

VIII. The Audit: Give, Take, or Refuse

Algorithm 1 states the procedure. It has three outcomes rather than two, and the third is a third of what it does: the constraint will give, it will take, or the population admits no answer and the audit says so.

Four ways to refuse, and each answers something. UNSWEEPABLE (lines 6–8) fires before any constraint is fitted. On a task generous enough that the trivial predictor is already close to the model's accuracy, no deployable classifier can be moved far enough to cross anything: LSAC refuses at an accuracy floor of 0.890 against a viable band of 0.056, and the test is arithmetic on the baseline. VOID (line 16) means the pool change spans under two points across the entire viable band — median 1.47 — which tells a deployer the direction question is moot on their system, not that it is unanswerable, and is arguably the most useful verdict here. NON-MONOTONE (lines 20–22) exists because bracketing assumes one rising sign change and that assumption is population-dependent: among the ten largest states swept, three are U-shaped, and a bracket step assuming monotonicity would return a confident wrong direction precisely there. And INDETERMINATE (lines 28–29) refuses when the deployed rate sits inside the located bracket or the nearest arms' effects are within seed noise — the two conditions under which this project's own record shows a sign call has nothing to grip.

Two cautions the earlier version kept in prose are now steps. Arms below a rate of 0.10 are marked advisory and excluded from the shape call (line 14), because below that boundary the two routes to a rate diverge and the accuracy guard does not catch it. And every fitted arm is probed for the flat keep-probability signature (line 19), because a lottery under the certificate is a finding the direction verdict alone would hide: flag an arm whose mean granted probability just below the boundary is under 0.10 while the keep-probability's correlation with the score above it is ≈ 0, a separation the data makes wide (0.00–0.07 against 0.17–0.43).

The magnitude guard: existence measured, value operational. Line 28's refusal rests on the claim that a near-zero effect has no sign to predict, and two sealed cohorts put a number on it: across forty arms the rule is correct on **21 of 22** above one point and **11 of 18** below — 95% against 61%, a coin flip — and clustering on the sample rather than the arm leaves the split intact. That measures the floor's *existence*. It does not locate it: sweeping the guard over the same arms, every value from 0.25 to 5.00 points separates by between 13 and 35 percentage points, with the largest separation at 0.25 and the committed 1.0 giving $+34$. The guard's value is a round number fixed in a protocol before its arms existed, which the data neither singles out nor contradicts.

What it returns, and what it does not. A sign, never a size: distance from the crossover orders the effect within a population, so a team can rank its own segments, but no slope transfers between populations. The audit also applies only where the pool can change size — where the allocation is capacity-pinned, a fixed number of seats or beds or slots, the total is constant by construction and the audit is out of

Algorithm 1 Directional audit: decide whether the predictor applies at all; if it does, locate the crossover and decide whether to add a floor

Require: deployed model h, held-out (X, y, a) *of the vintage to be deployed on*, fairness constraint C
Ensure: predicted direction, and whether a selection-rate floor is needed
1: **if** C does not constrain selection rates **then**
2:     **return** OUT OF SCOPE {e.g. equalized odds}
3: **end if**
4: $\pi \leftarrow \max(\bar{y},\, 1-\bar{y})$ {trivial-predictor accuracy}
5: $s \leftarrow h(X)$
6: $\mathcal{V} \leftarrow \{\tau : \mathrm{acc}(s>\tau) \geq \pi \ \wedge\ 0.05 \leq \overline{(s>\tau)} \leq 0.95\}$
7: **if** $\mathrm{span}(\mathcal{V}) < 0.40$ **then**
8:     **return** UNSWEEPABLE; compare natural sub-populations
9: **end if**
10: $T \leftarrow 12$ thresholds from $\mathcal{V}$, spread evenly in *selection rate*
11: **for all** $\tau \in T$ **do**
12:     $r_\tau \leftarrow \overline{(s>\tau)}$; fit C on the arm thresholded at τ
13:     $\Delta_\tau \leftarrow$ % change in favourable decisions
14:     discard τ if parity gap < 0.05 or $\mathrm{acc} < \pi$; mark τ *advisory* if $r_\tau < 0.10$ {route-divergence region}
15: **end for**
16: **if** fewer than 4 non-advisory arms remain, **or** $\max\Delta - \min\Delta < 2$ points **then**
17:     **return** VOID {nothing moved; not a refutation}
18: **end if**
19: probe each fitted arm's mixture for a flat keep-probability (Sec. IX-E); flag any flat arm
20: **if** $\mathrm{sign}(\Delta_\tau)$ changes more than once across the non-advisory arms, **or** changes from $+$ to $-$ as r_τ rises **then**
21:     **return** NON-MONOTONE {the rule does not apply here}
22: **end if**
23: **if** every non-advisory Δ_τ shares one sign **then**
24:     **return** that sign's verdict {it holds across the whole viable band}
25: **end if**
26: $c \leftarrow \big(\max\{r_\tau : \Delta_\tau < 0\},\ \min\{r_\tau : \Delta_\tau > 0\}\big)$
27: $r_0 \leftarrow$ selection rate of the deployed model
28: **if** $\min c < r_0 < \max c$, **or** $|\Delta|$ at the arms nearest r_0 is within seed noise **then**
29:     **return** INDETERMINATE {the sealed record's misses live here}
30: **end if**
31: **if** $r_0 \leq \min c$ **then**
32:     **return** WITHDRAWAL; add a selection-rate floor
33: **else**
34:     **return** EXTENSION; the floor buys little
35: **end if**

scope from line 0, which is the fixed-resource regime of [17]. No natural operating point below 0.13 appears in this record, so deployed rates beneath it carry an untested-region caution on top of any verdict.

What it needs and what it costs. A held-out sample carrying the protected attribute: audit-time access, which is legally and practically distinct from decision-time use and is the regime bias-testing provisions contemplate. It fits the constraint 12× seeds times; with five seeds and linear learners, one fit takes 2–10 seconds at 15k–69k training rows, so a full audit is minutes per population, scaling linearly in the base learner's own fit time since each reduction call is a sequence of learner refits. On a production gradient-boosted model over millions of rows that multiplier is the binding cost, and we have not measured it there.

What the sweep buys that one fit does not. A team able to run 12× seeds constrained fits could instead run *one*, at the operating point it already uses, and read the pool change off it — so what has prediction bought over measurement? A single fit returns a point; the sweep returns a neighbourhood. Across 639 adjacent arm pairs whose effects both clear the 1.0-point guard, 69 — 11% — disagree on the *sign*, so the direction one fit reports is not reliably a property of the population rather than of the threshold that fit used. And populations sit close to their own boundary: of 21 disjoint person samples carrying both a located crossover and a natural arm, 11 sit within 0.10 of the crossover and 6 within 0.05, Florida 2018 sitting 0.004 from it. For those teams the direction is not a fact to accept but a threshold they control, and one fit reports the withdrawal without reporting the lever. That the crossover cannot be looked up rather than measured shows in the same set: Florida enters twice, once per attribute, at 0.284 by sex and 0.439 by race.

The limit belongs with the answer. The sweep does not buy a better estimate of what the constraint does at today's threshold — repeated fits at one point buy that. It buys *shape*. Five of the 21 sit more than 0.20 from their crossover and would have decided the same from one fit; that 24% overstates the waste, because every sample in the set has already passed the monotonicity screen and the 22% admitting no rule never enter it. A team cannot learn which group it is in without running the sweep, which is the reply: the audit is not circular, because its output is not the quantity a single fit measures.

Two rates, and they are not the same rate. The survey's 78% and the audit's 29 of 52 are different measurements and a reader is entitled to be confused by them meeting on one page. The 78% applies *one* gate — monotonicity — to fifty populations drawn at random, and answers how often a population is shaped so that a directional rule could apply. The 29 of 52 applies *every* gate the algorithm states to the stored sweeps, which are a convenience set rather than a random draw, and answers how often the procedure returns a direction. The second is what a deploying team should plan against; the first bounds the claim.

The frame, since "at random" means nothing without one. Every combination of the fifty US states, the 2014, 2018 and 2019 ACS vintages and the two protected attributes, with the income cutoff held at $50,000 — 300 population-label pairs, from which fifty were drawn under a fixed seed. 2022 is excluded, for the diagnosed reason of Section XI. The frame, the seed, the draw and the exclusion were all committed to version control before any arm ran. Fifty draws resolve to 48 disjoint person samples, because two state-years were drawn under both attributes — the same people read twice.

The result: 78% admit a directional rule and 22% do not, of which 64% are classic single-crossing landscapes. Resampling populations rather than draws puts the directional share at 66–89%, and leaving each population out in turn never moves it below 77%. Run over the 52 stored sweeps that resolve to a dataset spec, the audit returns 29 directional verdicts, 8 NON-MONOTONE, 13 VOID, one refusal at the noise floor and one bracket with no natural arm — so it gives a direction a little under six times in ten, tells a team the question is moot or the rule inapplicable in most of the remainder, and is silent on four of 52. Every one of the 29 matches what the fitted constraint did at that population's natural operating point, which is a consistency check on the audit's reading of its own sweep rather than independent validation, since the natural arm is part of the curve being read.

The direction survives the extraction a deployer must apply. Every direction result here is measured on the randomized mixture, but a lender under adverse-action duties deploys a *deterministic* model. Thresholding the mixture's per-person probability at 0.5 on the natural arms of ten populations spanning both directions, the extracted model agrees with the mixture's pool direction on **10 of 10**, with extracted gaps of 0.009–0.034 against baseline gaps of 0.08–0.19. The caveat composes with the lottery: these are natural arms where the mixture is graded, and a flat-lottery arm has no score-respecting extraction — derandomising it undoes the parity it bought.

Guard provenance. A guard's provenance belongs to the pair it forms with a result, not to the guard. The band span, the parity-gap floor and the threshold grid are exploratory, fixed in the threshold sweep's own pre-registration and confirmatory for every seal after it. Three are not, and the complete record states which: a void guard written for a later pre-registration and imported *backwards*, moving three arm sets from refutation to VOID — a change in the conjecture's favour, against which the same audit found no arm set that passed on noise; an advisory boundary descending from a clause that was sealed, scored 4 of 8 and withdrawn, surviving only where it can refuse an arm but not manufacture a correct call; and a noise floor this paper's own drafting mis-dated, which is exploratory read from its value and retrospective read from the constant now carrying it.

IX. Boundaries

A. Where the crossover sits, and what it cost to find out

The crossover was first published as a cluster, on four populations spanning three instruments, three domains and two countries. It did not survive measurement. Two things about

TABLE VII

CROSSOVER LOCATIONS, WITH THE *existence* OF A CROSSING SEPARATED FROM ITS *location*. **P(CROSSING)** IS THE SHARE OF NESTED ROWS-BY-SEEDS RESAMPLES IN WHICH A SIGN CHANGE CAN BE BRACKETED AT ALL; **CROSSOVER** IS THE MEDIAN LOCATION CONDITIONAL ON ONE EXISTING. BRACKET-ONLY AND LENDING ROWS ARE LABELLED WHERE THEY ASSUME RATHER THAN ESTIMATE.

Population	**Domain**	**P(crossing)**	**Crossover *given* one**	**Uncertainty**
Nested rows-by-seeds bootstrap, 1,000 pairs				
COMPAS	crim. justice	100%	0.457	0.433–0.539
S. Carolina	income	97%	0.525	0.454–0.537
Oregon	income	99%	0.532	0.514–0.604
Dutch census	occupation	**0%**	—	none crosses
Bracket only — crossing assumed, not estimated				
Florida	income	—	**0.284**	0.259–0.309
Ohio	income	—	0.556	0.528–0.585
Pennsylvania	income	—	**0.652**	0.624–0.681
Lending — one market, three overlapping estimates				
HMDA pooled	mortgages	—	0.660	point only
HMDA Louisiana	mortgages	—	0.659	point only
HMDA refinance	mortgages	—	0.758	point only

Table VII before its numbers are read. Separating whether a crossing exists from where it sits is not pedantry: read under seeds alone the Dutch census appeared to cross at 0.576 with a 95% interval of 0.572–0.580, an apparently precise estimate, while under the nested bootstrap *no* resample brackets a crossing at all — the published figure was a property of the six arms that produced it. An interval conditioned on an event that may not occur cannot be read as an interval. And the rates in that table are not one economic object: survey rows are predicted-status rates over sampled persons, lending rows are approval rates conditional on having applied, and COMPAS inverts a risk score, so a crossover is comparable across them only as a position within each instrument's own range.

Located crossovers span 0.22 to 0.81 — 0.22 to 0.65 across the ACS income sweeps, with the law-school and South Carolina mortgage brackets above 0.80 — so the location is population-specific and only sometimes near the middle. Nothing in this work predicts it: the sealed test of the two candidate predictors returned UNDERPOWERED with three crossovers located against a floor of six, and three of ten large states have no crossover to locate at all, sweeping U-shaped or levelling up throughout. The lending estimates come from one two-state market read three overlapping ways, so "lending is different" remains a hypothesis rather than a finding.

That history is the case for measuring rather than transporting, and it is made by the prior's own failures rather than against them. Sweeping the arms where a fixed 0.54 fails shows they are populations with no directional rule to get right — non-monotone or outright inverted, including one the rule scored *correct*, whose correctness was therefore luck. Pooling every arm from all four sealed cohorts against its own cohort's crossover gives the calibration that motivates the INDETERMINATE verdict: the rule is correct on **19 of 19 arms more than 0.30 from the crossover, 23 of 27 between 0.15 and 0.30, 5 of 6 between 0.05 and 0.15 — and 1 of 6 within 0.05**. The last figure is not significant on six arms, but

TABLE VIII

SENSITIVITY TO THE PARITY-GAP EXCLUSION. CORRELATION PER POPULATION AS THE THRESHOLD IS LOOSENED.

Population	**0.01**	**0.02**	**0.05**	**0.10**
Dutch census	+0.851	+0.908	+0.946	+0.938
COMPAS	+0.844	+0.844	+0.844	+0.871
S. Carolina	+0.012	+0.012	+0.905	+0.849
Oregon	+0.095	+0.095	+0.664	—
Alabama	−0.368	−0.368	−0.368	+0.296
Kentucky	−0.590	−0.590	−0.654	—

TABLE IX

THE SEALED PREDICTION. POPULATIONS NEVER PREVIOUSLY MEASURED; PREDICTIONS COMMITTED BEFORE THE RUNS.

Population	**Rate**	**$\triangle$pool**	**Pred.**	**Actual**
MO, $100k	0.026	−16.48%	up	**down**
AZ, $100k	0.037	−23.34%	up	**down**
IN, $70k	0.081	−23.32%	up	**down**
TN, $50k	0.255	−1.68%	down	down
MD, $50k	0.508	+0.78%	down	**up**
MI, $30k	0.612	+0.85%	up	up
NC, $20k	0.789	+0.16%	up	up
GA, $10k	0.905	+0.30%	up	up
Taiwan 2005	0.885	+0.68%	up	up

a prior read off a dashboard cannot detect any of it, because detecting it requires the sweep the prior exists to avoid. The audit refuses all of them; only the prior is exposed.

B. A sealed prediction, and what it cost

The first sealed direction test failed: 4 of 8 against a bar of 7, not beating a constant (Table IX). What failed is a *refinement*, not the rule. Two hours before sealing we added a low-rate clause drawn from four in-sample populations — internally consistent, backed by a twenty-seed replication — and it is the clause that lost: scored on the same eight arms, the unrefined rule we held before the refinement gives seven of eight, its only miss being Maryland at 0.508, within 0.032 of the crossover on an effect of +0.78%. **That seven is post-hoc and we do not claim it; what was sealed scored four.** The clause was route-specific: its own arms reach a low rate by thresholding one fixed score vector, while the sealed arms reach it with a genuinely rarer label, and at that rate the two routes disagree completely — the clause held where it was fitted and failed on every label-route arm it was sealed against. Nothing in the in-sample data could have shown that. It is withdrawn, the unrefined rule was re-sealed on ten fresh populations, and the low-rate observation survives only in the weaker role of an exclusion (Section VIII), where it can refuse an arm but cannot manufacture a correct call.

C. The rule re-sealed, and it holds

The unrefined rule — down below 0.54, up at or above — was committed with its ten populations and its pass mark before any of their arms existed: at least 9 of 10 correct *and* strictly beating the best constant. **It holds: 9 of 10, best constant 6** (Table X). The re-seal applies none of this paper's

TABLE X

THE RE-SEALED PREDICTION: THE UNREFINED RULE — DOWN BELOW 0.54, UP AT OR ABOVE — COMMITTED BEFORE TEN NEVER-MEASURED STATES RAN. BAR: 9 OF 10 *and* BEAT THE BEST CONSTANT.

Population	Rate	Δpool	Pred.	Actual
NE, \$100k	0.014	-6.35%	down	down
AR, \$50k	0.195	-2.55%	down	down
OK, \$50k	0.220	-12.78%	down	down
WA, \$70k	0.233	-5.52%	down	down
NV, \$50k	0.297	-1.57%	down	down
MN, \$30k	0.699	-0.04%	up	**down**
CO, \$30k	0.701	$+0.63\%$	up	up
KS, \$20k	0.767	$+0.58\%$	up	up
WI, \$20k	0.814	$+0.69\%$	up	up
IA, \$10k	0.896	$+0.04\%$	up	up

tuned guards and scores all ten populations raw, so it has nowhere to hide an exclusion, which makes it the cleanest scoring here.

Two statistics, because they answer different questions and the weaker one is the honest headline. A guesser at the constant's rate reaches 9 or more with probability 0.046. But rule and constant disagree on only five arms, the rule wins four of those five, and the one-sided sign test on the discordant calls gives $p \approx 0.19$. Nor is this the first sealed attempt at a direction rule, so the binomial tail carries a sequential correction whose size is a judgement we should not make in our own favour: read narrowly it is a factor of two and the tail is near 0.09; read as this paper's own abstract counts them — twelve sealed or registered direction tests — it is a factor of twelve and the tail is 0.56. Across the conventional line either way, and not close to it on the wider reading.

One of the nine was luck, and the seed-level accounting says which. Sixteen of the nineteen sealed arms across both cohorts are unanimous in sign across their five seeds, and the three that split are exactly the near-zero effects. Minnesota, the miss, splits 3 of 5 — and **Iowa, at $+0.04\%$, splits 3 of 5 identically and scored correct**. Two statistically indistinguishable arms, one counted for the rule and one against it, which is what scoring near-zero magnitudes means and why Algorithm 1 now refuses them as INDETERMINATE.

What the pass cannot establish, stated where the score is. Every arm reaches its rate by the income-cutoff route, and the achieved rates leave the interior of $(0.30, 0.70)$ unsampled — the nearest sit at 0.297, 0.699 and 0.701. So the pass cannot separate the selection rate from label rarity as the operative variable, nor the value 0.54 from any prior in that interval: **a rule reading only the income cutoff scores identically here**. The screen-gated race cohort was registered to break exactly this confound and did, on mixed-route arms off-instrument and off-continent, where the cutoff-only null scored 5 of 10 against the rate rules' 8–9. Ten arms from two person-samples is not ten independent tests: the confound breaks *within* populations rather than across them, two clusters admit no usable bootstrap, and **leaving either population out drops the rate rule from 8 of 10 to 60%**. The result is real and its effects are large — to -47% — and its replication rests on two person-samples and should be read that way. Its sealed S1 nonetheless *fails* by its own letter, because a 0.50 prior edged the 0.54 prior 9 to 8: the rate carries the prediction and the specific value does not.

The paired statistic has a ceiling this design cannot raise. A third cohort was built to supply the discordant arms the sign test needs, and its pre-committed magnitude guard removed ten of twenty-four — removing them *near the crossover*, because effect size grows with distance from it ($r = +0.648$). Four discordant arms survived, the rule won all four, and four of four is $p = 0.0625$: the bar was arithmetically unreachable. The guard and the discordance a paired test needs are anti-correlated by construction, so the weakness does not yield to a larger balanced cohort. On a second, post-hoc cohort of larger states the fixed prior scored 5 of 10, every miss agreeing with that state's own measured crossover — and on a third, sealed on real mortgage approvals where the rule could at last be tested below a crossover, 5 of 10 again (Section XI). Three cohorts, one pass, which is the case for measuring rather than transporting, made by the prior's own failures.

D. Small populations

Below roughly 2,500 test subjects the method's own randomness exceeds the entire effect of the constraint. This is a specific instance of predictive multiplicity [21], [25] — and Long *et al.* [21] show fairness interventions themselves exacerbate it — models with equivalent aggregate performance disagreeing substantially on individuals — and we position it as a caution supported by that literature rather than as a novel finding. COMPAS has 1,584 and flips sign between seeds on two of four arms; LSAC has 6,240 and flips on none. COMPAS carries direction here and never magnitude.

E. How the withdrawal is implemented: a lottery

At severely cut operating points the fitted mixture stops being a graded adjustment and becomes a lottery: nobody below the threshold receives any probability, and every predicted positive of both groups is kept at one flat rate — 0.65 on the Dutch census arm, 0.135 on COMPAS — with keep probability uncorrelated with the person's own score. The gap closes by random discard and the certificate is unchanged. The charge here is deliberately narrower than "a lottery occurred": randomness has been advocated as a fairness device [24], the optimal aware classifier is itself randomized [22], derandomization is a studied problem [23], and on a claims-based view a lottery among relevantly equal claims can be what fairness requires [29]. The charge is that a lottery occurred *unannounced*, under a certificate indistinguishable from an informed adjustment, in a setting where individualized reasons are owed. **It is a checkable hazard rather than a cost of parity**: a control at each population's natural operating point finds every mixture graded, so the signature belongs to constraints pushed to extremes and not to withdrawal as such — which is why Algorithm 1 probes for it (line 19) and why the full treatment, including the search of a richer two-threshold class that returns an empty feasible set, is reported separately.

X. Method, and What We Got Wrong

Table XI is the complete ledger — every registered or sealed test, with its committed bar and its outcome, failures included and uncorrected. Counted once and consistently it holds **twenty-one** rows: fifteen fail, three record a hold (one alongside a failed first stage), two returned underpowered, and one settled a reading rather than a pass or a fail. Of the *twelve* that test a *direction* rule on populations it had not seen, one passed. The rows should not be read as twenty-one attempts at one claim of which three succeeded: each targeted a different claim, each failure removed a stronger claim than the one that survives, and the last column records what each row settled. The failures are boundary measurements; the passes are tests of the claim *as those boundaries define it*.

Three episodes are worth stating in the body because each cost a claim the paper would otherwise still be making. The rule does not survive equalized odds ($r = +0.334$ against a $+0.70$ bar), which is why the claim reads *parity* and not *group fairness*: parity controls approval rates directly and equalized odds does not. Our attempt to *derive* the rule was beaten by a constant, which is why the paper is empirical and says *predicts* rather than *causes*. And a refinement drawn from four in-sample populations, internally consistent and backed by a twenty-seed replication, was sealed, scored 4 of 8, and made the prediction worse than the unrefined rule it replaced — nothing in the in-sample data could have shown that, which is the whole argument for sealing.

Two published results of our own are withdrawn here rather than quietly corrected, and they failed for different reasons. Correlations previously reported at $+0.979$ and $+0.968$ are void because the operating-point design varies the rate by degrading the classifier and nothing bounded how far: excluding arms beaten by the trivial predictor leaves Alabama with two arms and LSAC with one. That same exclusion *rescued* mortgage lending, which had failed outright — one rule moving two results in opposite directions, which is why its provenance is stated in Section III. Separately, a post-processing comparison that appeared to show a clean reversal was arithmetic: the method re-derives its own thresholds and returned an identical model at all six operating points, so the effect was a constant divided by a shrinking baseline, and what exposed it was an exchange rate of exactly 0.000. A published crossover was also downgraded by a bootstrap we chose to run against our own result.

TABLE XI

The eight registered tests that impose a scope condition on the surviving claim. The ledger runs to twenty-one rows — fifteen fail, three hold, two underpowered, one a reading rather than a verdict — and the complete record carries all of them with their registration and scoring commits. Nothing is withdrawn here; ten rows are elsewhere.

Test	Bar / outcome	What it settled
EO transfer	$r \geq +0.70$; $+0.334$ — fail	the rule works for parity, which controls approval rates directly; it failed for equalized odds, which does not
Post-processing transfer	carry over; $r = -0.024$ — fail	the rule stops where the model may read the protected attribute — the boundary of Sec. VII
Derivation	beat a constant; beaten — fail	our own attempt to derive the rule lost to a constant; the first-order account we could not extend to the blind randomized case is named in Sec. II
Sealed rule, refined	7 of 8; got 4 — fail	the extra clause added from early data was wrong, and is withdrawn
Re-seal, unrefined rule	9 of 10 + beat constant; got 9, constant 6 — **holds**	the simple rule predicts populations it has never seen, from the rate alone
Sealed magnitude model	beat predict-zero; MAE 4.50 v. 0.77 — fail	size cannot be predicted from distance and band span; the direction-only concession is earned
Regime deconfounding	two sealed readings; R1 18/18, $r = +0.672$ — method reading	the pool boundary is the optimizer family, not blindness; the theorem's group directions hold in every aware cell (Sec. VII)
Race cohort (screen-gated)	S1: 8/10, but 0.5-null 9 — fail; S2 — holds	the cutoff null is beaten 8–9 vs 5: **it is the rate**, not label rarity; the 0.54 value holds no in-band privilege; within-population passes off-instrument at $\rho \geq 0.9$
Sealed allocation cohort	12/16 + beat 3 nulls; rule 7/8, constant 7/8 — fail	HMDA 2021 at purpose level, sealed to reach below the crossover: **0 of 8 arms sit below it**, so a constant ties the rule. Third demonstration that US mortgage approval cannot test the rule; the rate does beat the purpose-only null 7–6
Dwelling cohort (7.1b)	6/8 + beat 3 nulls; rule 5/8, constant 6/8 — fail	the transported 0.660 is badly placed for manufactured housing: on ten such arms the pool change is monotone in the rate with *one* sign change, bracketing a crossover at 0.39–0.42. The prior scores 5 of 10 there and a crossover measured on the same arms 10 of 10 — post-hoc, and not claimed. First real-allocation arms below any crossover
Located crossover (7.2)	beat transported + constant; 1/3 v. 3/3 — fail	0.405, located on ten manufactured markets, loses to the transported 0.660 on the three fresh markets that separate them — and both misses sit inside 0.405's own dead band. Over all sixteen arms 0.405 calls 13 and 0.660 calls 9, but that is post-hoc

XI. Limitations

The lending evidence is thinner than its market count suggests. Forty markets carry a rate-bearing arm on each attribute, but every one of the twenty-six sex arms falls below the audit's own disparity floor — median baseline gap 0.013 against the race arms' 0.079 — so the allocation result rests entirely on race arms. That is a property of US mortgage approval rather than a gap in the method, and the audit refuses those arms correctly. Nor does either lending count discriminate: every arm sits at a rate of 0.82 or above, so the rule predicts extension on all of them and a constant scores identically. **US mortgage approval supplies no market whose rate straddles a crossover, and that is now established three ways rather than inferred once.** The state-level arms sit at 0.82 and above; the fifty-market coverage sits at 0.82 and above; and a sealed cohort built specifically to reach lower — a reporting year this work had never touched, split by loan purpose because purpose is what moves the approval rate — returned **0 of 8 arms below the crossover** and failed for exactly that reason, the rule scoring 7 of 8 against a constant's 7 of 8 (Section X). On real approvals the rule and "always extend" are the same rule. Testing the direction claim on allocations needs a register whose accept rates reach below roughly 0.66, which is a statement about what is publicly available rather than about the rule.

The one place the rule could finally be tested on real approvals, and what happened there. Every lending test above failed for the same non-reason: US mortgage approval sits above the crossover, so the rule predicts extension everywhere and cannot be told apart from a constant. One slice of the register reaches lower — *manufactured housing*, approved at 0.505 against site-built's 0.850 — and a cohort pairing both slices in each of seven markets was sealed on it, with a void condition committed in advance for the case where the rate rule and a dwelling-only null never separate. **The sealed test fails**: the rule transported at 0.660 scores 5 of 8 against a constant's 6 of 8, though it does beat the dwelling null. **On ten such arms the signs sorted; on sixteen they do not, and the difference is the whole lesson.** The first ten manufactured arms ran monotone with a single sign change, bracketing a crossover near 0.405, and we reported that as the phenomenon holding. Six further markets, sealed against that located value, give five sign changes and the located rule loses to the transported one on the arms that separate them. **We withdraw the monotonicity reading**: this paper's monotonicity claim is *within* a population across operating points, and ten populations compared at their natural rates is a different object — one this paper elsewhere says does not transport. What survives is a cross-population correlation that is now significant where it was not, $\rho = +0.747$ ($p = 0.001$, $n = 16$), and the fact that no single threshold separates these markets: the best possible calls 14 of 16, the located 0.405 calls 13, and the paper's own lending value of 0.660 calls 9 — worse than a constant's 10. Five sealed attempts on real approvals have now failed in five different ways, and what none of them has established is that *any* transported crossover predicts direction on allocative decisions.

Magnitude does not transfer between populations, only within them, and a sealed model of it lost to predicting zero. **The crossover's location is unexplained and wider than it looked**: located values span 0.22 to 0.81, the sealed test of the two candidate predictors returned underpowered, and some populations have no crossover to locate. **We cannot say why the rule works.** Our own derivation lost to a constant; the first-order account we could not extend to the blind randomized case is named in Section II.

A label is not a task. A fixed nominal outcome definition is a different task each vintage: a dollar cutoff slides down an inflated income distribution, and restoring the earlier shape requires matching the label's *quantile*, not its real price — CPI-anchoring is not enough. Four 2022 populations invert for this reason, and one 2014 state inverts at a lower cutoff, so this is a property of nominal labels rather than of a vintage.

Coverage is eight data sources, five countries and two decades, of which 170 of 179 populations come from two instruments; the eight sources remain the honest count of independent instruments. Taiwan, Brazil and Mexico are the non-Western populations and arrived late enough that most of the record predates them. The only EU population is the 2001 Dutch census, which by this paper's own vintage standard grounds no claim about current EU deployments.

XII. Discussion

What a practitioner can use. Run the audit on your own data. It will locate your crossover, tell you your system admits no directional rule, or tell you the question is moot because the pool barely moves across your whole viable band — and each of those is an answer. Where it returns a direction and that direction is withdrawal, a selection-rate floor is the remedy: not a new method, but a linear moment constraint inside the reductions framework [1] and a variant of the minimum-rate constraints of [11]. Table XII is what it buys, measured on the branch the audit actually sends it to — the **95 natural arms across 78 populations** that clear both of the audit's gates and on which the plain constraint withdraws. It costs **0.05 accuracy points** on average, moves the median exchange rate from 1.32 destroyed per created to 0.94, and brings **78 of the 95** below one-for-one, none of which were below it already. On the median arm it does not merely soften the withdrawal but reverses it, from -2.78% to $+0.97\%$.

Three things that should be said with those numbers rather than after them. The count is over arms and the populations are 78, so the ratio is not 95 independent trials. Sub-1.0 is a descriptive count of predicted labels and not a welfare verdict, per Section IV-A — the group-conditional flows are what a weighted reading would use. And earlier versions of this paper reported the floor's benefit as scaling with the damage at $r \approx -0.99$; we withdraw that, because the benefit is the damage minus the remainder and the two share a term, so the correlation is arithmetic rather than a finding. Two honest consequences follow. Because the floor is that cheap, a team that expects withdrawal and does not want to sweep should simply apply it; what the audit adds is the ability to tell in advance whether it is needed, how far the operating point sits from the sign change, and whether the population admits an answer at all. And whatever the audit returns, the deployed system's pool change and who-pays decomposition should be measured afterwards and reported beside the parity number — so that escaping the ex-ante audit buys no escape from the accounting.

What to report. The parity difference alone certifies both outcomes this paper distinguishes. A report that adds the pool change and the group-conditional flows costs nothing to compute, requires no doctrinal commitment, and is the difference between a dashboard that can tell those outcomes apart and one that cannot.

Adversarial inputs. A procedure that can refuse can also be gamed, in four places we can name: the label definition, the segmentation the audit is run over, the sweep's own

TABLE XII
WHAT THE SELECTION-RATE FLOOR BUYS, ON THE 95 NATURAL ARMS OVER 78 POPULATIONS THAT CLEAR THE AUDIT'S NOISE AND PARITY GATES AND ON WHICH THE PLAIN CONSTRAINT WITHDRAWS. MEDIANS EXCEPT WHERE STATED; EVERY FIGURE RECOMPUTED FROM STORED ARMS.

	plain	**with floor**
Exchange rate (destroyed per created)	1.32	**0.94**
Arms at or below one-for-one	0 of 95	**78 of 95**
Change in the pool	-2.78%	$+\mathbf{0.97}\%$
Accuracy, mean cost of the floor	—	0.05 pts

parameters, and refusal used as an exit. The countermeasures are symmetrical — fix the label's quantile rather than its nominal value, pre-register the segmentation, freeze the sweep grid, and pair every verdict including a refusal with the ex-post accounting above. One concrete instance is worth stating: on HMDA a single constraint earns "extension" on the pooled book while concentrating withdrawal in home improvement, so a segmentation chosen after the fact can hide the thing the audit exists to surface.

Appendix: Two Worked Traces of Algorithm 1

Oregon 2018, sex, to a verdict. Demographic parity constrains selection rates: in scope (line 1). Trivial-predictor accuracy $\pi = 0.636$ (line 4); the viable band clears the 0.40 span bar, so the sweep proceeds (lines 6–9). The stored sweep holds 18 arms; the guards discard 2 on the parity-gap floor and 2 as worse than doing nothing, leaving 14 spanning rates 0.111–0.653, none below the 0.10 advisory boundary (lines 11–15). Their pool changes span 7.14 points, clearing the void guard (line 16). The signs, sorted by rate, are $-----------+++$: one rising change, so the landscape is monotone-crossing (lines 20–25) and the bracket is $c = (0.528, 0.589)$ (line 26). The deployed model's rate is $r_0 = 0.353$ — below $\min c$, outside seed noise: verdict WITHDRAWAL, add a floor (line 32). What the constraint in fact does at Oregon's natural arm: -3.0% of the pool.

LSAC bar passage, to a refusal. Trivial-predictor accuracy $\pi = 0.890$ — almost everyone passes. The viable band spans 0.056, far under 0.40: verdict UNSWEEPABLE at line 8, before a single constraint is fitted. Had the guard been ignored, the stored sweep shows what awaited: five of seven arms fall below the trivial predictor and the two survivors are too few to read — VOID. Both endings refuse; neither manufactures a direction.

Reproducibility

Code, stored results and the full document trail are in the accompanying repository, which holds the per-test registration and scoring commits, the OpenTimestamps receipts for the seals that carry them, and the complete twenty-one-row ledger. Every figure here is re-derived from stored results by a test suite that fails if a documented number and its experiment disagree, and counts appearing in more than one place are derived from one source and checked against each other — after an audit found three descriptions of the same table. One limit cannot be repaired retroactively: for ten of the sixteen seals the ordering rests on git commit timestamps from our own machine, which an outsider must take on trust. The later seals carry external anchoring; the earlier ones do not.

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